

Bayesian Prior Elicitation for Mechanistic Tumor Evolution Models: A Comprehensive Review of Parameters and Calibration Dynamics

The quantification of tumor evolution through mechanistic models requires the meticulous specification of parameters governing cellular proliferation, chromosomal instability, whole-genome doubling, and allometric scaling. In a continuous-time multitype branching process or ordinary differential equation framework, these parameters dictate the topological evolution of the cellular state space and the macroscopic manifestation of tumor burden. The elicitation of Bayesian priors for such models is a complex and highly nuanced exercise. Overly restrictive informative priors may bias the posterior away from the true data-generating mechanism, while entirely uninformative or flat priors often lead to pathological posterior geometries, unidentifiable parameter ridges, and non-convergent Hamiltonian Monte Carlo sampling trajectories. Furthermore, because mechanistic models bridge the microscopic scale of single-cell cytogenetics with the macroscopic scale of clinical tumor volume, parameterization inherently suffers from ill-posed inverse problems where disparate biological mechanisms can produce mathematically identical macroscopic outputs.

This comprehensive report provides an exhaustive analysis of five critical parameters within a mechanistic ploidy-evolution and tumor-burden model: the base missegregation probability scale (μ), the whole-genome doubling branch probability scale (δ), the minimum proliferation rate (λ_{min}), the positive proliferation-rate gap (λ_{gap}), and the ploidy-to-cell-volume scaling exponent (η). The analysis systematically synthesizes empirical biological evidence, explicitly converts disparate measurement scales into a unified mathematical modeling framework, and addresses severe identifiability constraints. Based on these foundations, the report constructs robust, weakly informative candidate priors. The ultimate objective of these priors is to regularize the inference process, ensuring that the model strictly adheres to physical and biological reality while retaining sufficient flexibility to allow the observed clinical and genomic data to drive the posterior distribution.

Base Missegregation Probability Scale (μ)

Biological Mechanisms and Empirical Constraints

The parameter μ , frequently denoted in the literature as p_{misseg} , is strictly defined within the

context of the model as the per-chromosome missegregation probability per cell division. It serves as the fundamental base scale for the missegregation kernel, which subsequently determines the probability distribution of the net chromosome shift during any given cellular division event. Chromosomal instability, which is quantified by this parameter, is a pervasive hallmark of cancer that fuels intra-tumor heterogeneity, facilitates evolutionary adaptation, and mediates mechanisms of therapeutic resistance. The molecular etiology of chromosomal instability encompasses a variety of structural and functional defects, prominently including merotelic kinetochore-microtubule attachments, the transient formation of multipolar spindles, and dysfunction within the anaphase-promoting complex.¹

Empirical measurements of chromosome missegregation rates are highly sensitive to the experimental technique employed, the baseline ploidy of the observed cells, and the specific definitions of error rates reported in primary literature. In normal, euploid human cells, chromosome missegregation is an exceptionally rare event. Fluorescence in situ hybridization and live-cell imaging studies indicate that missegregation occurs at a rate of approximately 1.2% to 2.3% per mitosis in healthy cell populations.³ However, this metric represents the aggregate probability of *any* missegregation occurring across the entire genome during a single cell division. To align with the mechanistic parameter μ , this per-cell or per-mitosis rate must be mathematically transformed into a strictly per-chromosome rate.

Assuming that chromosome missegregation events are independent and identically distributed across the genome, the probability of a cell experiencing at least one

missegregation event (P_{cell_error}) is related to the per-chromosome missegregation

probability (μ) and the total number of chromosomes (N) by the equation

$P_{cell_error} = 1 - (1 - \mu)^N$. For very small values of μ , this relationship can be closely

approximated by the linear expansion $P_{cell_error} \approx N \times \mu$. Given a standard diploid human karyotype of $N = 46$ chromosomes and an empirical per-cell error rate of approximately 1%,

the baseline per-chromosome probability is derived as $\mu \approx 0.01/46$, which yields a value on the order of 2.17×10^{-4} .¹

In neoplastic populations, the missegregation rate is elevated by one to two orders of magnitude compared to healthy tissue.³ Direct observations of cancer-derived cell lines exhibiting severe chromosomal instability reveal per-chromosome missegregation rates ranging from 0.3% to 1.0% per division.⁵ For example, studies assessing aneuploid colorectal cancer cell lines indicate that these cells missegregate a chromosome every cell division or every few divisions, translating to a per-chromosome rate ranging from approximately 1.3×10^{-3} to 1.0×10^{-2} .⁸ Furthermore, computational models utilizing the CINner mathematical framework for pan-cancer analysis have established plausible search spaces

and prior distributions for the missegregation probability p_{misseg} bounding it broadly between 10^{-4} and 10^{-1} , and slightly more narrowly for non-whole-genome-doubled cells between 10^{-4} and 10^{-2} .

A critical secondary insight governing the elicitation of priors for μ is the concept of a density-dependent evolutionary limit, often referred to as error catastrophe. Mathematical models of intra-tumor heterogeneity and spatial growth indicate that there is an absolute upper limit to the missegregation rate, beyond which the tumor population experiences unsustainable genomic degradation, leading to population collapse and extinction.³ This boundary is intricately tied to the cellular turnover rate of the tissue. While highly unstable cells might theoretically reach per-chromosome error rates approaching 5% (0.05) under extreme experimental perturbations, such as acute exposure to the MPS1 inhibitor Reversine¹, sustained naturally occurring rates in established, viable tumors rarely exceed the 10^{-2} threshold.

Construction of Weakly Informative Priors for μ

Given that μ is a dimensionless probability, it is mathematically required to be strictly bounded within the interval $[0, 1]$. The elicitation of a weakly informative prior must reflect the empirical boundaries established above, heavily concentrating probability mass in the 10^{-4} to 10^{-2} regime, while strictly preventing the inference engine from exploring biologically lethal spaces approaching 1.0.

The first candidate is a Beta distribution parameterized by shape and scale parameters. The Beta family serves as the natural conjugate prior for probabilities defined on the standard unit interval. A weakly informative formulation is $\mu \sim \text{Beta}(1.5, 300)$. This specific parameterization yields an expected mean of approximately 0.005 (equivalent to a 0.5% error rate per chromosome) and features a broad variance that easily encompasses the plausible literature range. Crucially, selecting a shape parameter $\alpha = 1.5$ (which is strictly greater than 1) ensures that the probability density function approaches zero at exactly $\mu = 0$. This avoids the pathological singularity or "pole" at zero that occurs when $\alpha < 1$, which can severely disrupt the leapfrog integration steps of a Hamiltonian Monte Carlo sampler. The β parameter of 300 strongly shrinks the upper probability mass, perfectly reflecting the biological reality of error catastrophe by rendering values above 0.05 highly improbable.

The second candidate is a truncated Log-Normal distribution. Because missegregation probabilities span multiple orders of magnitude across different tumor phenotypes, modeling

the parameter on a strictly positive logarithmic scale is highly effective. A suitable formulation is $\mu \sim \text{LogNormal}(-5.5, 1.5)$ truncated to the interval $[\delta, 1.0]$. A location parameter of -5.5 on the log scale corresponds to a median probability of exactly 0.004 . The scale parameter $\sigma = 1.5$ provides a wide 95% credible interval that cleanly spans from 10^{-4} up to 0.05 , capturing the full spectrum of both low-instability and high-instability neoplasms. The hard truncation at 1.0 ensures physical validity, though the probability mass naturally extending above 1.0 under this parameterization is already negligible.

The third candidate is a Logit-Normal distribution applied to a transformed parameter space. For optimal geometric exploration by advanced gradient-based samplers like the No-U-Turn Sampler, transforming constrained probabilities to an unconstrained real-number line via the logit function is often superior to using bounded distributions. The parameter is defined as $\mu_{\text{logit}} = \text{logit}(\mu) = \ln(\mu/(1 - \mu))$, with the prior formulated as $\mu_{\text{logit}} \sim \text{Normal}(-5.0, 1.5)$. This approach places the median expected probability at the inverse-logit of -5.0 , which is approximately 0.006 . The $\pm 2\sigma$ bounds cleanly encapsulate the 10^{-4} to 10^{-1} interval. By avoiding hard boundaries entirely, this prior facilitates exceptionally smooth posterior exploration, making it the most robust choice for complex, high-dimensional mechanistic models.

Whole-Genome Doubling Probability Scale (δ)

Biological Mechanisms and Empirical Constraints

The parameter δ , frequently denoted as p_{wgd} in the literature, is defined as the scalar probability of a whole-genome doubling event occurring during any single cellular division event. It dictates the transition rates within the whole-genome doubling branching matrices of the mechanistic model. Whole-genome doubling is a profound macro-evolutionary leap in tumorigenesis, providing neoplastic cells with an entirely duplicated set of chromosomes. This polyploidization can occur through several well-documented mechanisms, including cytokinesis failure, endoreduplication, mitotic slippage, or the physical fusion of adjacent cells.¹¹

Biologically, the immediate aftermath of a whole-genome doubling event imposes a severe fitness penalty on the newly formed tetraploid cell. The massive increase in cellular and nuclear volume, coupled with supernumerary centrosomes, typically induces prolonged cell cycle arrest, cellular senescence, or apoptosis driven by intact p53 surveillance pathways and the Hippo signaling cascade.¹³ Consequently, the successful propagation of a whole-genome doubled lineage almost universally requires preexisting or concurrent somatic mutations in

tumor suppressor genes, particularly TP53.¹⁵ However, if the tetraploid cell manages to bypass these restrictive checkpoints, the duplicated genome provides a profound evolutionary advantage. The redundancy of the genetic code buffers the cell against the deleterious effects of subsequent chromosomal missegregations and loss-of-heterozygosity events, allowing the lineage to navigate tight evolutionary bottlenecks and tolerate a state of hyper-aneuploidy that would be lethal to a diploid progenitor.⁸

Because the whole-genome doubled state provides such a potent buffering capacity against genomic decay, its ultimate manifestation is observed at exceptionally high prevalence rates in cross-sectional genomic data. Comprehensive pan-cancer analyses demonstrate that approximately 30% of all advanced human solid tumors exhibit genomic signatures indicative of a historical whole-genome doubling event, with certain lineages like lung adenocarcinoma and colorectal cancer exhibiting even higher frequencies.¹⁵

However, a critical distinction must be maintained between the historical *prevalence* of whole-genome doubling in a mature tumor and the actual *rate* of whole-genome doubling per cell division (δ). The probability of a whole-genome doubling event occurring and surviving during any single mitosis is exceedingly low. If δ were relatively high (for instance, greater than 0.05), the tumor population would rapidly and continually cascade into unsustainable levels of extreme polyploidy (e.g., octaploidy or hexadecaploidy), which is rarely observed in human clinical samples outside of specialized tissues like megakaryocytes or syncytiotrophoblasts.¹³ Quantitative empirical estimates of the per-division whole-genome doubling rate are notoriously sparse due to the technical difficulties of observing rare, transient macro-mutational events in real-time. In sophisticated mathematical models designed to reconstruct tumor phylogenies and estimate these rates from bulk whole-genome sequencing data—such as the CINner framework—the prior distribution for the whole-genome doubling probability (p_{wgd}) is explicitly set to span the ultra-rare interval between 10^{-6} and 10^{-3} .⁹ In another study utilizing an agent-based simulation of B-cell acute lymphoblastic leukemia, the probability of successfully undergoing whole-genome doubling was calibrated to a fixed constant of 0.011 day^{-1} .¹⁷ However, this specific value aggregates both the baseline occurrence of the event and the immediate survival probability of the resulting polyploid clone within a highly accelerated computational environment, representing a theoretical upper limit rather than a physiological baseline. Theoretical consensus dictates that δ must be strictly lower than the base individual chromosome missegregation rate μ , positioning it firmly in the 10^{-6} to 10^{-4} regime for the vast majority of somatic mammalian cells.

Construction of Weakly Informative Priors for δ

Like μ , the parameter δ is a dimensionless probability and requires mathematical constraints strictly bounding it within the interval $[[0, 1]]$. Because δ represents a catastrophic, macro-

evolutionary saltation that is expected to be infinitesimally small on a per-division basis, the chosen prior must exert heavy, regularizing shrinkage toward zero. This prevents the inference engine from erroneously attributing standard, incremental aneuploidy to continuous cycles of whole-genome doubling.

The first candidate is a Beta distribution parameterized to heavily favor values near zero. The formulation $\delta \sim \text{Beta}(1.0, 10000)$ is mathematically ideal for this purpose. By setting the

shape parameter $\alpha = 1.0$, the resulting probability density function is strictly monotonically decreasing and reaches its absolute maximum peak exactly at zero. This represents the strongest possible prior belief that, in the absence of overwhelming data, a cell will *not*

undergo whole-genome doubling. A scale parameter $\beta = 10000$ provides an expected mean of exactly 10^{-4} and an extremely sharp decay profile. This serves as a powerful regularizing prior, ensuring the model only infers a non-zero whole-genome doubling rate when the genomic sequence data displays an undeniable, ubiquitous duplication of multiple chromosome arms across the inferred phylogeny.

The second candidate is a truncated Log-Normal distribution. The formulation

$\delta \sim \text{LogNormal}(-9.2, 2.0)$ truncated to the interval $[\delta_{\min}, \delta_{\max}]$ directly models the phenomenon across logarithmic scales. A location parameter of -9.2 on the natural log scale corresponds to a median probability of exactly 1.0×10^{-4} . The standard deviation of $\sigma = 2.0$ is

intentionally specified to be wider than the scale used for the missegregation probability μ .

This increased variance reflects the higher degree of epistemic uncertainty regarding in vivo whole-genome doubling rates, providing adequate support for probabilities as astonishingly low as 10^{-6} while maintaining the capacity to reach 10^{-2} if necessitated by the likelihood.

The third candidate relies on the logit transformation to remove hard boundaries, defined as

$\delta_{\text{logit}} = \text{logit}(\delta)$. The prior is formulated as $\delta_{\text{logit}} \sim \text{Normal}(-9.0, 2.0)$. This

parameterization maps the extreme rarity of the whole-genome doubling event into a smooth, continuous, unbounded parameter space. When modeling rare, catastrophic biological events, Bayesian posterior distributions often exhibit pathological geometric features such as "funnels" near the zero boundary. The logit-normal formulation aids the Hamiltonian Monte Carlo algorithm in efficiently exploring these complex posterior geometries without encountering the divergent transitions that frequently plague bounded parameter spaces.

Minimum Proliferation Rate ($\lambda_{\min}$)

Biological Mechanisms and Empirical Constraints

The parameter $\lambda_{\min}$ represents the absolute minimum baseline cell division rate for the

division-rate function, acting as the mathematical floor that populates the diagonal division-rate matrices within the continuous-time Markov model. The units for this parameter are defined as day^{-1} . To rigorously triangulate biologically plausible values for λ_{mir} , one must evaluate the macroscopic doubling times of clinical tumors and deconstruct them into the microscopic durations of the eukaryotic cell cycle.

Clinical observations utilizing advanced radiographic imaging, thymidine labeling indices, and flow cytometry indicate that macroscopic human tumor doubling times (T_d) vary dramatically across different anatomical sites and histological grades. These macroscopic doubling times range from less than one week in highly aggressive lymphomas to over one year in indolent neuroendocrine tumors, with a broad pan-cancer median value of approximately 60 days.¹⁹ However, it is a fundamental principle of tumor kinetics that the macroscopic doubling time is a function of the net growth rate, which is the mathematical difference between the cellular proliferation rate and the cellular death rate.

The true, underlying duration of the cell cycle (T_c) for actively dividing human tumor cells is surprisingly conserved across cancer types. The mean cell cycle duration for proliferating tumor cells is roughly 2 days, with relatively minor variations between different solid tumors.¹⁹

This temporal duration can be explicitly converted into a theoretical division rate (λ). In an idealized exponential growth model without cell death, the population size $N(t) = N_0 e^{\lambda t}$.

The doubling time is defined as $T_d = \ln(2)/\lambda$. Therefore, the division rate is calculated as $\lambda = \ln(2)/T_c \approx 0.693/T_c$. A cell cycle duration of 2 days equates to a division rate of approximately 0.346 day^{-1} .²⁰

However, intra-tumor heterogeneity ensures that not all cells within a neoplasm cycle at this idealized, maximum rate. A substantial proportion of cells within any solid tumor may enter a quiescent G_0 state due to nutrient deprivation, hypoxia, or intrinsic phenotypic plasticity, resulting in a significantly lower effective division rate for certain sub-populations.²¹

Experimental data on breast cancer models and acute lymphoblastic leukemia indicate that effective division rates are distributed across wide intervals, frequently spanning $[0.2, 0.4] \text{ day}^{-1}$.¹⁷ In elegant spatial studies measuring division strictly at the extracellular matrix and microvessel interface, tumor cell proliferation was recorded at roughly 0.016 hour^{-1} , which explicitly converts to approximately 0.38 day^{-1} .²²

The absolute minimum division rate, λ_{mir} , corresponds to the slowest-cycling cells that are still considered actively participating in the progression of the tumor. This expressly excludes wholly dormant cells, senescent fractions, or necrotic tissue that no longer contributes to the transition matrices of the model. A theoretical floor corresponding to a cell cycle duration ranging from 10 days to 50 days is biologically realistic for slow-cycling, drug-tolerant persister

cells. Using the standard conversion formula $\lambda = 0.693/T_c$, a 50-day cell cycle yields a rate of $\approx 0.014 \text{ day}^{-1}$, and a 10-day cell cycle yields a rate of $\approx 0.069 \text{ day}^{-1}$. Therefore, empirical evidence strongly suggests that λ_{mir} should realistically be bounded within the interval of 0.02 to 0.1 day^{-1} .

Construction of Weakly Informative Priors for λ_{mir}

Because λ_{mir} represents a strict rate of occurrence over time, it must be bounded as a strictly positive real number (> 0). Weakly informative priors must penalize values approaching exactly zero to avoid catastrophic computational singularities—such as infinite division times crashing the ordinary differential equation solver—while remaining sufficiently dispersed to cover the empirical variability of slow-cycling cellular sub-populations.

The first candidate is a Gamma distribution, which is the mathematically conjugate prior for Poisson and exponential rate parameters and is natively restricted to the positive real domain $(0, \infty)$. The formulation $\lambda_{min} \sim \text{Gamma}(2.0, 0.05)$ parameterized by shape (k) and scale (θ) is highly appropriate. This parameterization yields an expected mean of 0.10 day^{-1} and a distinct mode at 0.05 day^{-1} , which precisely corresponds to a conservative 14-day cell cycle. Crucially, the shape parameter $k = 2.0$ ensures that the probability density drops to exactly zero at the origin. This topological feature naturally pushes the inference model away from a biologically implausible state of absolute cellular stasis, while the scale parameter remains flexible enough to allow very small positive values if demanded by the longitudinal tumor burden data.

The second candidate is a Log-Normal distribution, formulated as

$\lambda_{min} \sim \text{LogNormal}(-3.0, 1.0)$. A location parameter of -3.0 on the natural logarithmic scale places the median proliferation rate at approximately 0.05 day^{-1} . The standard deviation of $\sigma = 1.0$ generates a 95% credible interval spanning roughly 0.007 day^{-1} to 0.35 day^{-1} . This prior geometry perfectly encapsulates the vast uncertainty regarding the absolute proliferation floor without asserting a strict, artificial upper boundary, allowing the model to smoothly accommodate both indolent and highly aggressive neoplastic phenotypes.

The third candidate is a Half-Normal distribution with an explicit positive offset, defined as $\lambda_{min} \sim \text{Normal}(0.05, 0.1)$ truncated to the interval $[0.001, \infty]$. This is a simple, computationally highly efficient prior that concentrates probability mass near the biological expectation of 0.05 . The hard lower truncation at 0.001 mathematically enforces a maximum

allowable cell cycle duration of nearly 700 days. While technically a hard boundary, it is placed so far outside the realm of active cellular biology that it acts merely as a computational safeguard, preventing the exploration of pathological near-zero values that might otherwise destabilize the transition rate matrices of the multitype branching process.

Positive Proliferation-Rate Gap (λ_{gap})

Biological Mechanisms and Empirical Constraints

The parameter λ_{gap} mathematically defines the positive gap between the maximum allowable division rate and the minimum baseline division rate within the tumor architecture. It is formally established by the equation $\lambda_{max} = \lambda_{min} + \lambda_{gap}$. In the context of mechanistic modeling, this parameter encapsulates the phenotypic plasticity and intra-tumor temporal heterogeneity of cell cycle durations, dictating how rapidly the fastest subclones can outcompete their slower counterparts.

As established in the preceding section, the minimum possible cell cycle duration in mammalian biology is physically and biochemically constrained. The time required for accurate DNA synthesis (S-phase), coupled with the complex mechanical choreography of chromosomal alignment and segregation during mitosis (M-phase), institutes an absolute

biological speed limit on cellular division. Typical eukaryotic cell cycle times (T_c) rarely, if ever, fall below 24 hours in human somatic cells.¹⁹ Therefore, the absolute maximum physiological division rate (λ_{max}) is bounded near 0.693 day^{-1} (assuming a 24-hour cycle) or potentially up to 1.0 day^{-1} under conditions of extreme oncogenic hyper-proliferation.²⁰

Given that the minimum baseline rate (λ_{min}) typically resides within the interval of 0.02 to 0.1 day^{-1} , the gap parameter (λ_{gap}) must absorb the mathematical remainder of the scale to reach the physiological ceiling of λ_{max} . Consequently, simple arithmetic constraints dictate that λ_{gap} is biologically confined to lie within the interval of 0.2 to 0.9 day^{-1} . Experimental and clinical modeling robustly corroborates this derivation. In complex mathematical models simulating continuous tumor growth and chemotherapeutic intervention, parameters characterizing the division rate frequently converge around 0.2 to 0.4 day^{-1} .¹⁷ A gap parameter significantly larger than 1.5 day^{-1} would imply the existence of a cellular subset capable of completing a full division cycle every 11 hours or faster. While such hyper-proliferation is occasionally observed in highly aggressive hematopoietic malignancies, embryonic stem cells, or specialized in vitro cell lines, it is largely anomalous and physiologically unsustainable for the solid mammalian tumors typically evaluated by these mechanistic models.²⁰

Construction of Weakly Informative Priors for λ_{gap}

The parameter λ_{gap} must be strictly positive to satisfy the logical requirement that $\lambda_{max} > \lambda_{min}$. An optimal weakly informative prior for this parameter should heavily favor the empirically validated 0.2 to 0.8 range, while utilizing a rapid exponential decay in the upper tail to aggressively suppress mathematically valid but physically impossible hyper-proliferation regimes.

The first candidate is a Gamma distribution, formulated as $\lambda_{gap} \sim \text{Gamma}(4.0, 0.125)$. This specific parameterization yields an expected mathematical mean of 0.5 day^{-1} and a distinct mode of 0.375 day^{-1} . The variance is tightly constrained by the shape parameter of 4.0, ensuring that the cumulative probability of λ_{gap} exceeding 1.5 is astronomically low. This sophisticated prior beautifully encodes the physiological speed limit of mammalian cell division, gently guiding the inference algorithm away from hyper-proliferation without imposing the strict, artificial hard thresholds that can cause gradient evaluations to fail. The second candidate is a standard Truncated Normal distribution, defined as $\lambda_{gap} \sim \text{Normal}(0.5, 0.3)$ truncated to the interval $[0, \infty]$. This classical approach centers the proliferation gap directly at the expected biological mean, representing a 2-day variance in cell cycle times across the intra-tumor hierarchy. The standard deviation of 0.3 is sufficiently wide and diffuse to allow the likelihood function to pull the gap closer to 0.1 or up toward 1.0, completely dependent upon the specific velocity of the macroscopic tumor burden trajectory observed in the clinical data.

The third candidate is a Log-Normal distribution, formulated as $\lambda_{gap} \sim \text{LogNormal}(-0.7, 0.5)$. The inherent properties of the log-normal family natively ensure strict positivity. A location parameter of -0.7 equates to a median gap of approximately 0.49 day^{-1} . The relatively tight scale parameter ($\sigma = 0.5$) ensures that the distribution does not suffer from a heavily dispersed right tail. This is computationally vital, as it strictly prevents the Hamiltonian optimizer from wandering into unstable, unphysical hyper-proliferation spaces during the erratic early warm-up iterations of the Markov Chain Monte Carlo sampling phase.

Ploidy-to-Cell-Volume Scaling Exponent (η)

Biological Mechanisms and Empirical Constraints

The parameter η serves as the critical allometric scaling exponent that bridges the

unobservable, microscopic genomic state space (ploidy) to the macroscopic, physically measurable clinical observable (cell volume, which aggregates into total tumor burden). The concept of allometric scaling between DNA content and cell size is a foundational, albeit highly debated, principle of theoretical cell biology. It is traditionally formalized by the power-law function: $\text{Volume} \propto \text{Ploidy}^\eta$.²⁴

Historical and contemporary biological consensus posits a strict linear relationship between nuclear DNA content and cellular volume across many eukaryotic organisms, implying an expected scaling exponent of exactly $\eta = 1.0$. Compelling evidence from model organisms such as budding and fission yeast, as well as cross-species analyses of mammalian erythrocytes, demonstrates nearly perfect linear scaling. This suggests that the raw mass of chromosomal material proportionally dictates the size of the cell in order to maintain a rigid, homeostatic cytoplasm-to-genome ratio.²⁴

However, when this principle is empirically tested against highly heterogeneous human cancer cell lines, the linear scaling hypothesis frequently reveals profound nuances and contradictions. A rigorous, systematic analysis conducted across the extensive NCI60 cancer cell line panel identified that volumetric growth, ribosomal translation initiation, and overall cellular metabolism display a $\approx 2/3$ allometric scaling exponent against DNA content. The precise statistical estimate was $\eta \approx 0.66$, accompanied by a tight 95% confidence interval ranging from 0.61 to 0.71.²⁶ This $2/3$ exponent is not an arbitrary mathematical artifact; it is deeply rooted in the biophysical "Pump-Leak" model and the unyielding physical constraints of geometry. Cellular metabolism and nutrient uptake are rate-limited by the surface area of the cell membrane, which scales at exactly a $2/3$ power relative to cellular volume. Consequently, waste elimination decreases less rapidly than nutrient uptake as a cell grows larger, forcing the cell to cap its volumetric expansion relative to its ploidy.²⁶

Furthermore, the linear scaling hypothesis is known to break down entirely following catastrophic whole-genome doubling events. Post-WGD tetraploid colorectal cancer cells (such as the DLD1 line) exhibit staggering levels of both nuclear and cellular size heterogeneity, completely uncoupling the strict 1:1 ploidy-to-volume ratio observed in euploid cells.³⁰ Extreme states of aneuploidy can also result in severe cytoplasm dilution, causing scaling laws to naturally fail due to the biochemical saturation of RNA polymerases and the complex osmotic balance dictated by chromatin counterions.²⁴

Consequently, an epistemological conflict exists in the current literature: basic homeostatic principles suggest $\eta = 1.0$, whereas metabolic constraints, surface-area limits, and direct measurements of cancer cell lines suggest $\eta \approx 0.66$ to 0.75. Therefore, the prior for η must be carefully centered between these two competing biological realities, remaining sufficiently broad to allow the specific tumor dataset to select the optimal geometric map.

Construction of Weakly Informative Priors for η

Because η is an exponent that determines a continuous scaling function mapping ploidy to volume, it is theoretically unbounded on the real number line, but practically restricted to positive values near 1.0. An exponent less than 0 would imply the biological absurdity that cells shrink as they gain DNA, and an exponent greater than 1.5 would imply a physically unsustainable geometric expansion that violates basic conservation of mass.

The first candidate is a standard Normal distribution, defined as $\eta \sim \text{Normal}(0.85, 0.15)$.

This represents an exemplary, state-of-the-art weakly informative prior. The mean of 0.85 elegantly splits the difference between the $2/3$ (0.66) metabolic and surface-area scaling limit²⁶ and the 1.0 linear DNA-content scaling limit.²⁴ The standard deviation of 0.15 allocates approximately 68% of the probability mass exactly between 0.70 and 1.0, accommodating both physical theories equally while heavily penalizing extreme values less than 0.4 or greater than 1.3.

The second candidate is a Truncated Normal distribution, formulated as

$\eta \sim \text{Normal}(1.0, 0.2)$ truncated to the strict interval $[0.5, 1.5]$. This specific prior

formulation fundamentally assumes the null hypothesis of pure linear scaling ($\mu = 1.0$)²⁴, but provides a wide enough variance to allow the likelihood function to correct the exponent downward toward 0.66 if surface-area metabolic limits are dominating the growth of the specific tumor type being modeled. The hard bounds prevent devastating numerical overflow errors in the volume-mapping function during the integration of the ordinary differential equations.

The third candidate is a Log-Normal distribution, formulated as

$\eta \sim \text{LogNormal}(-0.15, 0.25)$. The utilization of a log-normal prior ensures strict positivity, entirely eliminating the risk of negative exponents. The location parameter of $\nu = -0.15$ places the median near 0.86, aligning harmoniously with the literature synthesis.

The asymmetrical, heavier right tail of the log-normal distribution permits occasional hyper-scaling events without forcing them. This topological feature is exceptionally ideal for modeling advanced tumors characterized by substantial WGD-induced size heterogeneity, where cytoplasm dilution may cause unpredictable volumetric ballooning.³⁰

Calibration-Aware Recommendations and Identifiability Constraints

When attempting to calibrate a continuous-time multitype branching model of tumor evolution

using only longitudinal tumor burden data and terminal ploidy assessments, the parameter space is inherently beset by severe structural and practical unidentifiability. Weakly informative priors, functioning as mathematical regularizers, are absolutely imperative to tame the geometry of the posterior distribution and prevent the sampler from becoming hopelessly trapped in non-identifiable parameter ridges.³¹ The interplay between these five parameters creates complex multidimensional correlations that must be carefully managed.

Resolving Confounding Between Incremental CIN and Catastrophic WGD

The parameters governing chromosomal instability (μ) and whole-genome doubling (δ) are highly prone to profound confounding if the sole inferential target is the terminal ploidy distribution of the tumor.⁸ A tumor exhibiting a high frequency of near-tetraploid or hyper-triploid cells could have theoretically arrived at this complex state through two entirely disparate mechanisms: a single, rare macro-evolutionary jump (high δ) followed by minor stabilization, or through the continuous, incremental accumulation of individual chromosomes driven by a hyper-elevated missegregation rate over a long period (high μ). To ensure mathematical identifiability and biological fidelity, the priors for these two parameters must be structurally distinct and asymmetrical. The whole-genome doubling probability δ must be anchored with an aggressively tight prior favoring extremely rare occurrence rates (e.g., 10^{-4}), strictly reflecting its status as a catastrophic, bottleneck-inducing event. Conversely, the base missegregation probability μ can tolerate a significantly broader prior. By imposing a highly restrictive prior on δ , the model is fundamentally forced to explain generic, low-level aneuploidy via μ . It will only invoke the whole-genome doubling parameter when the terminal genomic data showcases an undeniable, simultaneous duplication of multiple chromosome arms across the inferred phylogeny.³⁵ Stress-testing this dynamic involves running prior predictive checks: artificially widening the prior on δ will almost certainly cause a precipitous, correlated drop in the posterior estimate of μ , explicitly highlighting the direct trade-off embedded within the likelihood space.

Decoupling Net Tumor Burden from Absolute Proliferation Rates

In mathematical oncology, utilizing macroscopic tumor burden—measured in volume or aggregate cell counts—to infer microscopic, single-cell division rates presents a classic, notoriously ill-posed inverse problem.³⁶ The observed macroscopic tumor volume is exclusively a function of net growth (cellular proliferation minus cellular death and immune clearance). Consequently, the absolute scale of the division rates (λ_{mir} and λ_{gap}) is weakly identifiable at best, because an infinite set of high-proliferation/high-death rate combinations can produce the exact same net tumor growth curve as a low-proliferation/low-death rate

combination.

Therefore, the priors placed on λ_{mir} and λ_{gap} must be informative enough to restrict the model strictly to biologically plausible cell cycle times (ranging between 1 and 50 days).¹⁹ If uninformative, flat priors are utilized, the Hamiltonian optimizer will inevitably drift into mathematically equivalent but biologically nonsensical parameter spaces—such as assigning a cell cycle of 5 minutes paired with a catastrophic, near-instantaneous death rate. A tighter, biologically grounded prior on λ_{gap} serves as the essential temporal anchor for the entire continuous-time Markov chain. By fixing the biological clock, it allows the mutation and instability parameters (μ and δ) to scale accurately relative to a realistic timeline of tumor evolution.

The Non-Linear Jacobian Influence of Allometric Scaling

The allometric scaling exponent η uniquely and disproportionately influences the likelihood function because it non-linearly maps the hidden, multidimensional ploidy state directly to the singular observed volume data. Small perturbations in the value of η exponentiate errors in the predicted tumor burden, potentially overwhelming the gradients of the other four parameters and dominating the trajectory of the MCMC sampler.

A narrow, well-calibrated prior on η (centered exactly between the theoretical 0.66 and 1.0 boundaries) is absolutely essential for stable calibration of the entire system. To rigorously stress-test the model's sensitivity, analysts should perform a series of computational experiments: fix η entirely at 1.0 (representing pure linear scaling) and subsequently at 0.66 (representing strict metabolic scaling) in separate parallel Markov chains. If the Maximum a Posteriori estimates for the instability parameter μ and the proliferation gap λ_{gap} fluctuate wildly between these two rigid states, it conclusively demonstrates that the model is overly sensitive to the volume mapping function. This diagnostic result indicates that η requires even stronger prior shrinkage to maintain global calibration and prevent the misinterpretation of tumor burden data.

Structured Summary of Parameter Evidence and Candidate Priors

Parameter	Definition (exact)	Units	Evidence-based plausible range (Derivation)	Candidate priors (2–3 options) with hyperparameters	Key citations

μ	Base missegregation probability scale (per chromosome).	Dimensionless probability (0–1).	10^{-4} to 10^{-2} : Normal cells exhibit $\sim$ per chromosome; established CIN tumor cell lines range from 0.3% to 1.0% (10^{-3} to 10^{-2}). An upper mathematical cap is required to prevent error catastrophe and population collapse.	1. Beta: $\alpha = 1.5, \beta =$ 2. LogNormal (Trunc): $\nu = -5.5, \sigma =$ 3. Logit-Normal: $\nu = -5.0, \sigma =$	1
δ	Scalar probability of whole-genome doubling per division event.	Dimensionless probability (0–1).	10^{-6} to 10^{-3} : While WGD is historically prevalent in cross-sectional genomic data (~30%), it is exceedingly rare on a per-division basis. Pan-cancer CINner framework modeling establishes priors strictly within this ultra-rare lower boundary.	1. Beta: $\alpha = 1.0, \beta =$ 2. LogNormal (Trunc): $\nu = -9.2, \sigma =$ 3. Logit-Normal: $\nu = -9.0, \sigma =$	9

λ_{mir}	Minimum cell division rate (a floor) for the division-rate function.	day^{-1}	0.02 to 0.1. Macroscopic tumor doubling time median is ~60 days. Persister and slow-cycling cells may take 10-50 days to complete a cell cycle, mathematically establishing the biological division floor.	1. Gamma: $k = 2.0, \theta =$ 2. LogNormal: $\nu = -3.0, \sigma =$ 3. Half-Normal (Trunc): $\mu = 0.05, \sigma =$	17
λ_{gap}	Positive gap between maximum and minimum division rates.	day^{-1}	0.2 to 0.9. Maximum cell cycle speed in human solid tumors is rarely faster than 1-2 days ($\lambda \approx 0.5 -$). Subtracting the baseline floor gives a gap that accurately captures intra-tumor division heterogeneity.	1. Gamma: $k = 4.0, \theta =$ 2. Trunc Normal: $\mu = 0.5, \sigma =$ 3. LogNormal: $\nu = -0.7, \sigma =$	17
η	Exponent linking ploidy to per-cell volume ($V \propto$).	Dimensionless	0.66 to 1.0. Metabolic surface-area constraints enforce $2/3$ scaling (0.66) in NCI60 cell	1. Normal: $\mu = 0.85, \sigma =$ 2. Trunc Normal: $\mu = 1.0, \sigma =$ 3. LogNormal:	24

			lines, whereas classic biological homeostatic models rigidly assume linear 1.0 scaling.	$\nu = -0.15, \sigma$	
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Final Prior Recommendations

The fundamental objective of eliciting these specific priors is to inject necessary biological realism into the mechanistic architecture without overwhelming the likelihood generated by the longitudinal data.

For the dimensionless probabilities representing mutational transitions, the Logit-Normal priors are definitively the most highly recommended. Implementing $\mu_{logit} \sim \text{Normal}(-5.0, 1.5)$ and $\delta_{logit} \sim \text{Normal}(-9.0, 2.0)$ provides the Hamiltonian Monte Carlo sampler with an unconstrained, smooth gradient geometry. Crucially, it strictly encodes the biological reality that whole-genome doubling is exponentially rarer than single-chromosome missegregation, mitigating parameter confounding and improving the structural identifiability of the ploidy topology.

For the temporal proliferation rates, Gamma distributions are the optimal choice because they strictly enforce positivity and gently slope toward zero, naturally preventing computational artifacts driven by hyper-proliferation. Initiating the model with $\lambda_{min} \sim \text{Gamma}(2.0, 0.05)$ and $\lambda_{gap} \sim \text{Gamma}(4.0, 0.125)$ safely anchors the simulation's temporal scale to a realistic 2-to-20 day cell cycle spectrum. This anchor is absolutely critical for resolving the mathematical unidentifiability between cellular division and cellular death rates that remains deeply hidden within aggregate bulk tumor burden data. Finally, for the allometric scaling exponent, the standard Normal prior formulated as $\text{Normal}(0.85, 0.15)$ stands as the most robust starting point. It brilliantly balances the empirical observation of $2/3$ metabolic surface-area scaling against the theoretical expectation of 1.0 linear DNA-volume scaling. By avoiding hard truncations on this specific exponent, the underlying optimization algorithm can seamlessly adjust the geometric map based directly on whether the specific clinical dataset exhibits significant, whole-genome-doubling-induced size heterogeneity.

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